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# ANDREA CAPPELLUTI

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## EDUCATION

**Master's Degree, LUISS Guido Carli** **Rome, Italy**
***Attending: Data Science and Management*** *(ENG)* 09/2024 – 10/2026

***Relevant Courses:***

Advanced Statistics (29/30), Econometrics (30L/30), Machine Learning (28/30), Data Privacy and Security (27/30), Data Visualization (28/30), Technology Entrepreneurship (30L/30).

**Thesis Title:** Forecasting Passenger Check-in Show-up Curves at Rome Fiumicino Airport: A Machine Learning Approach.

**Thesis Subject:** Machine Learning – Thesis Advisor: Giuseppe F. Italiano – Co-Advisor: Vincenzo Lomonaco

**Thesis Summary:** Forecasting system for passenger arrivals at check-in desks to support staffing decisions, built on 29.5M check-in records linked to 520K departures over 3+ years (5- to 30-minute bins, horizons from 1 hour to 1 week). A LightGBM multi-horizon model with leakage-free features (show-up kernel convolved with the flight schedule) cut error by 29–33% vs. the seasonal baseline, statistically significant on all setups. Benchmarked against XGBoost, CatBoost, deep learning models (TiDE, Temporal Fusion Transformer) trained on AWS SageMaker GPUs, and a LoRA fine-tuned TimesFM 3.0 time-series foundation model.

**Erasmus+ program, TalTech University** **Tallinn, Estonia**
09/2025 – 02/2026

**Bachelor's Degree, Università degli Studi Roma Tre** **Rome, Italy**
***Economics e Big Data*** *(ITA)* *09/2021 – 07/2024*

***Final Grade:*** *102/110*

***Relevant Courses:***

Databases and Big Data (29/30), Statistics for Big Data (26/30), Statistical Learning (30/30), Machine Learning and Data Processing (28/30), Digitalization of the Labor Market (30/30), Economics of Digital Markets (28/30)

**Thesis Title:** Implementation of Artificial Intelligence in Online Platform: Impacts, Opportunities and Challenges.

**Thesis Subject:** Economics of Digital Markets – Thesis Advisor: Eleonora Pierucci

## WORK EXPERIENCE

***AI Engineer, Yellow Tech - The AI Transformation Company*** *08/2026 – Present*

1. Design and build AI agents and develop AI-powered software solutions, delivering practical, end-to-end implementations for real production environments.
2. Optimize business processes and workflows by integrating AI-driven tools and automation, bringing intelligent automation into real production workflows.
3. Build document-processing pipelines that combine OCR models and LLMs to turn business documents into validated, structured data.

***Customer Insights Analyst, MSX International*** *04/2026 – 07/2026*

1. Maintain and process large-scale vehicle databases through data cleaning and quality control, optimizing operational workflows and minimizing service downtimes.
2. Analyze customer data to identify needs and deliver tailored solutions to Ford customers, leveraging data-driven insights to enhance customer satisfaction and service quality.

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***Data Science Intern, Basco&T Consulting*** *02/2026 – 04/2026*

1. Contributed to the development of Mobility Lens, an integrated analysis and visualization platform combining public transport data with real estate pricing across Milan, designed to translate complex datasets into clear and actionable strategic insights.
2. Built data pipelines and interactive dashboards to visualize mobility patterns and real estate market trends, enabling data-driven decision-making and supporting the delivery of strategic reports to clients in the public transport sector.
3. Built a RAG system to provide personalized reports and evaluations in the real estate market, including information about public transport and composite indicators of the neighborhood.

## OTHER CURRICULAR ACTIVITIES

***Multi-Agent System for Natural Language Analysis on Public Datasets – Reply Project*** *03/2025 – 04/2025*

1. Developed an AI-powered multi-agent system to transform natural language questions into automated data analysis and visualizations, using large language models (GPT-4.1), Python, and Streamlit.
2. Engineered an architecture using OpenAI's Agent SDK and LlamaIndex to interpret user queries, select appropriate public datasets, and dynamically generate and execute Python code (pandas, matplotlib).
3. Delivered a system that returns both narrative insights and graphical outputs, designed for scalable, self-service civic analytics to democratize access to complex data.

***Sales Forecasting System for Retail Analytics – Unieuro Project*** *03/2025 – 04/2025*

1. Led a data science project to predict 2024 sales volumes for key product categories (smartphones, vacuum cleaners) by combining business intelligence, statistical modeling, and machine learning.
2. Conducted extensive data preprocessing, data cleaning, and feature engineering to extract relevant signals such as seasonality, promotions, and historical demand trends.
3. Developed interactive dashboards using Power BI and compared multiple machine learning models (XGBoost, Gradient Boosting, Random Forest) to deliver actionable sales forecasts.

## SKILLS, ACTIVITIES & INTERESTS

**Languages:** Italian: Native | English: C1 | Spanish: A2

**IT Skills:** Python, R, SQL, Stata, DuckDB, Power BI, Tableau, Orange Data Mining, MS Office Suite

**Machine Learning:** scikit-learn, LightGBM, XGBoost, CatBoost, PyTorch, TensorFlow, Darts, Optuna, time-series forecasting, fine-tuning of foundation models (LoRA / PEFT)

**AI Skills:** AI agents and multi-agent systems, LLM APIs, System One decision models, Model Context Protocol (MCP), Retrieval Augmented Generation (RAG), OCR and LLM-based document data extraction, prompt engineering, Claude Code, Gemini CLI, Google AI Studio

**Cloud & Deployment:** Docker, Railway, Vercel, Supabase, AWS SageMaker (GPU training), Git

**Certificates:**

- **Anthropic:** Claude Certified Architect – Foundations (2026), Introduction to Model Context Protocol, AI Fluency: Framework & Foundations
- **Codecademy:** How to Analyze Business Metrics with SQL, Learn Statistics with R, BI Dashboards with Power BI, BI Dashboards with Tableau, Learn Prompt Engineering
- **LUISS Guido Carli:** AI Literacy Digital Certificate
- **StudySection E-Learning:** Python (Advanced)

**Activities and Interests:**

- Former Pro Basketball player
- Hiking and trekking photographer
- Sport science enthusiast

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